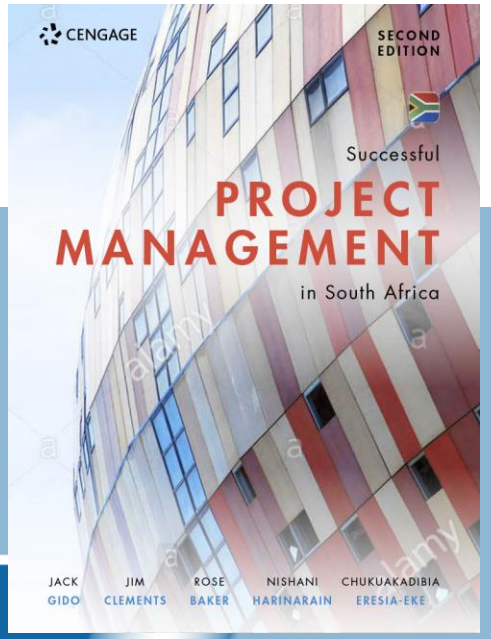
CHAPTER 8



Managing Risk

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Chapter Concepts

- Identifying risks and their potential impact
- Assessing the likelihood of occurrence and degree of impact of risks
- Risk response planning
- Controlling risks

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Learning Outcomes

- Discuss what is involved in managing risks
- Identify and categorise risks
- Assess and prioritise risks
- Prepare a risk response plan
- Develop a risk assessment matrix
- Control risks

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Performance Domains of PMBOK® 7th edition covered in this chapter:	Knowledge Areas of PMBOK® 6th edition covered in this chapter:
• Stakeholders • Team • Planning • Uncertainty	• Project Risk Management

Project Management Principles relevant to this chapter

Principle 7 (Tailor Based on Context)
Principle 9 (Navigating Complexity)
Principle 10 (Optimising Risk Responses)

Principle 10 – Risk: Uncertain events known as risks can have either a positive or negative impact on the project and need to be continually addressed throughout the life cycle of the project. How the risk is handled depends on the organisation's attitude, appetite and threshold for risk. The response to the risk occurrence needs to be realistic, appropriate, cost effective, owned by a relevant person and agreed to by the relevant stakeholders.

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- What is the **purpose** of **risk management**?
- Why is it **important** to **manage risks throughout** the **lifecycle** of a project?
- How could risks be **identified**, **categorised** and **assessed**?
- How should a **risk response plan** be **prepared**?
- How should a **risk assessment matrix** be developed?
- How could the project **risk control process** be described?
  - How should a **risk response plan** be **implemented**?
  - What does risk **monitoring** involve?

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Identify Risks

- Can jeopardise achieving the project objective
- Brainstorm sources of risks
- Establish categories for risks and evaluate
- Use historical information
- Progressively elaborate and identify new risks as more information becomes available

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Approaches to identifying risks

- Brainstorming
  - A common approach to identifying the sources of risks
  - The project manager should involve key project team members in identifying potential sources of risk
  - Each member of the project team can bring his or her experience and insight to help develop a comprehensive list of risks
  - Common sense and reason must prevail when identifying risks
- Risk Categories
  - Are groupings of potential sources of risk by topic and then identifying risks that might occur for each category
  - Examples of risk categories (pg 296 – 297):

a) Technical
b) Schedule
c) Cost
d) Human resources
e) External
f) Sponsor/customer

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Approaches to identifying risks

- Historical information

- Historical information from past projects can be helpful in identifying possible risks
- If post project evaluations are done on completed projects, they could be a good source for identifying possible risks
- Additional examples of specific risks – page 297

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Assess Risks

- Determine the likelihood the risk event will occur
- Evaluate degree of impact on the project objective
- Prioritise
  - Likelihood of occurrence and degree of impact
  - Position relative to the critical path

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Risk assessment matrix

FIGURE 8.1 Risk Assessment Matrix for an Outdoor Concert

RISK	IMPACT	LIKELIHOOD OF OCCURRENCE (L, M, H)	DEGREE OF IMPACT (L, M, H)	ACTION ON TRIGGER	RESPONSIBILITY	RESPONSE PLAN
Rain on day of event	• Low attendance • Incur financial loss	M	H	Weather forecast two days before event	Owethu	• Reserve indoor space now • Recruit extra volunteers to work around the clock to set up indoors • Develop detailed plan
Road construction	• Reduced attendance • Reduced revenue	H	H	Highway Dept. publishes construction schedule	Nothile	• Identify alternate routes • Have signs made • Post signs along all routes • Announce in news media

L = Low; M = Medium; H = High

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Plan Risk Responses

- Set of actions
  - Prevent or reduce the likelihood of occurrence or the impact of a risk
  - Implement if the risk event occurs
- Establishes a trigger point for implementing an action
- Assigns responsibility for implementation
- Avoid, mitigate, or accept the risk
- Include a contingency fund to cover implementation cost

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Control Risks

- Risk control includes implementing risk response plans and monitoring risks
- Risk response plans should be implemented as appropriate when their trigger point is reached
- Risk monitoring includes regularly reviewing the risk assessment matrix throughout the project
- Regularly review and evaluate all risks to determine whether there are any changes to the likelihood or potential impact of any of the risks
- Project meetings are a good forum for regularly reviewing, updating and addressing risks
- Track and document which risks actually occurred and their impact

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- What are **examples** of **IT project risks**?
  - How is **project risk managed** in these **projects**?
- What **principles and techniques** are involved in **managing risks** in **agile projects**?
- What are the **risk responses** in agile projects?
  - How would you explain the risk response **strategies** that are used **agile projects**?
- In what way can **IT** project teams in particular, **benefit** from **agile risk management techniques**?

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Managing Risks for Information Systems Development

- Risks can be categorised into seven types
  - Technological risk
  - Human risk
  - Usability risk
  - Project team risk
  - Project risk
  - Organisational risk
  - Strategic and political risk

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IS example: risk assessment matrix

FIGURE 8.2 Risk Assessment Matrix for Web-Based Reporting System Project

RISK	IMPACT	LIKELIHOOD OF OCCURRENCE (L, M, H)	DEGREE OF IMPACT (L, M, H)	ACTION ON TRIGGER	RESPONSIBILITY	RESPONSE PLAN
Lack of cooperation and commitment from users	Incorrect sales records in reporting system	M	H	Sales staff have difficulty using system during training	Jim	Have additional training materials to describe how to use
High number of ad hoc queries	Systems design not completed on time	L	H	More queries to be answered than time remaining to complete	Johan	Have assigned staff work longer on tasks; staff have at least seven days of slack
Testing reveals design problems	Rework the design and development	M	H	Evaluation reveals problems	Cathy	Examine for design flaws during Evaluation task
Customised software has integration problem	System not working and no information recorded	L	H	Study of existing system identifies potential problems	Fani	Develop system with compatible programming, possibly not latest technology
Changes in membership of the project team	Lack of proper skills and knowledge to do task	L	M	Resignation statement by staff member	Beth	Assign another project team member to learn task during notice period

L = Low; M = Medium; H = High

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Critical Success Factors

- **Identify risks** and their **potential impacts** before the project starts.
- **Involve** the **project team** or experts in assessing risks.
- **Assign high priority** to managing risks that have a high likelihood of occurrence and a high potential impact on the project outcome.
- **Develop response plans** for addressing high priority risks.

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Summary

- Risk is an uncertain event that, if it occurs, can jeopardise achieving the project objective.
- Risk management includes identification, assessment, control, and response to project risks in order to minimise the likelihood of occurrence and/or the potential impact of adverse events on the achievement of the project objective.
- Risk identification includes determining which risks may adversely affect the project objective and estimating what the potential impacts of each risk might be if it occurs.
- Assessing each risk involves determining the likelihood that the risk event will occur and the degree of impact the event will have on the project objective, and then prioritising the risks.
- A risk response plan is a defined set of actions to prevent or reduce the likelihood of occurrence or the impact of a risk, or to implement if the risk event occurs.
- Regularly review and evaluate all risks to determine if there are any changes to the likelihood of occurrence or the potential impact of any of the risks, or if any new risks have been identified.

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